EXPLORATORY METHODS BRIDGE · NOT A VALIDATED MODEL

The missing bridge is an observation model

A simulation predicts hidden states; flow cytometry measures a gated sample of cell-level outcomes.

1 · FORWARD MODEL

OpenFOAM: flow, shear and residence time

PHREEQC: aqueous speciation and mineral equilibria

Julia: cell-state, reaction and transport dynamics

2 · OBSERVATION OPERATOR

Project latent state to sample-specific outputs:

- absolute gated-cell concentration
- membrane-integrity gate fractions
- fluorescence / scatter distributions

Include calibration, sampling and recovery.

3 · INFERENCE & FALSIFICATION

Compare predicted and measured summaries.

Estimate uncertainty in rates.

Test rival mechanisms and held-out time points.

Do not treat one good fit as unique causality.

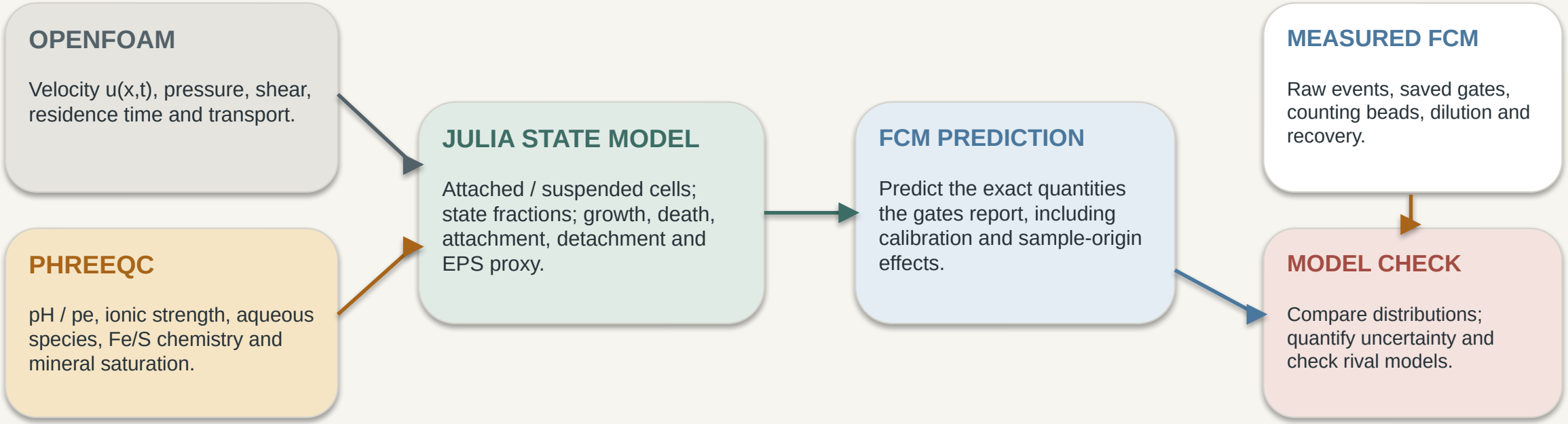
Flow cytometry belongs in the measurement / likelihood layer, not inside the CFD or geochemical solver.

Calibration precedent: Yip et al. (2022), PLOS Comput Biol 18(10):e1010533, https://doi.org/10.1371/journal.pcbi.1010533

ARCHITECTURE · KEEP THE SOLVERS IN THEIR LANES

One modular stack, two independent kinds of inputs

Only couple a solver if its fields represent a real part of the target experiment.



FCM loses spatial position: pair cell-state distributions with microscopy and endpoint surface / mineral analysis.

Nano-Bio Lab research overview: https://www.nanobiolab.org/research · Model-calibration review: https://doi.org/10.1371/journal.pcbi.1010533

SEPARATE APPLICATIONS · SHARE THE SOFTWARE PATTERN, NOT THE PARAMETERS

Two possible flow-cytometry branches should stay separate

The same observation-model idea can serve different biology, but the targets and calibration differ.

A · NANO-BIO / CELL-ASSOCIATION ASSAY

Possible observables:

- percent fluorescent-positive cells
- per-cell fluorescence distribution
- validated cell-state / membrane-integrity markers
- scatter / morphology proxies

Key control: distinguish surface association from internalization; cytometry alone may not localize the signal.

B · MICROBIAL / CLAY-BIOFILM ASSAY

Possible observables:

- absolute gated-cell concentration
- membrane-integrity state fractions
- detached vs planktonic sample differences
- optional validated metabolic panel

Key control: clay-only blanks, disaggregation recovery, staining controls and oxygen-safe sampling.

Do not merge the hypotheses: a nanocarrier uptake model is not the clay–corrosion model. Re-estimate chemistry, kinetics, gates and parameters for each system.

Flow-cytometric biofilm methods: Kerstens et al. (2015), https://doi.org/10.1007/s12223-015-0400-4; Grainha et al. (2020), https://doi.org/10.3390/antibiotics9110741

SMALLEST FALSIFIABLE WORKFLOW · DISCUSSION DRAFT

Make the cytometry link testable before fitting the model

The goal is to distinguish mechanisms, not to make a visual match look like a validated prediction.

1 · PREDECLARE

Sample origin, time points, stains, gates, instrument settings, primary summaries and exclusion rules.

2 · CALIBRATE

Unstained / stain-only, single-stain or compensation controls, live-like / dead-like controls, blank matrix and counting beads.

3 · MATCH THE STATE

Map each sample to effluent, planktonic or detached-biofilm model state. Record extraction recovery and dilution.

4 · TEST PREDICTION

Fit on designated observations; check posterior predictive distributions or held-out time points. Report non-identifiable alternatives.

Falsifier: if predicted cell-state distributions do not track the measurements under the predeclared calibration, revise or reject the state model; do not tune gates after seeing the desired result.

Assay cautions: Kerstens et al. (2015); Grainha et al. (2020). Calibration concepts: Yip et al. (2022).